29 July 2026

Editor
Electric Power Systems Research

Dear Editor,

Please consider our manuscript, "GFNO-PINO: A Topology-Disjoint Benchmark Isolating Physics Fine-Tuning, Topology Conditioning, and Spectral Filtering in AC-OPF Graph Surrogates," for publication in *Electric Power Systems Research*.

This is a controlled ablation and benchmark study. Its central finding is that physics-informed fine-tuning substantially improves the power-balance accuracy of graph-based AC-OPF surrogates under the executed protocol, whereas explicit topology conditioning and spectral graph filtering—architecturally plausible choices commonly assumed useful—are not confirmed to improve accuracy in matched-parameter, topology-disjoint comparisons. We report this negative result explicitly because it directly informs surrogate-model design choices.

Across five training seeds and four IEEE systems, paired physics fine-tuning reduces mean sample-maximum power-balance mismatch by 63.4–89.3%, with all 20 paired seed effects positive. Parameter-matched topology-blind and spatial-GNN baselines nevertheless match or outperform the full model on three of four systems.

We do not present the model as a production AC-OPF solver. Every learned model has 0% feasibility under the declared proxy, and the manuscript keeps this limitation prominent. It also distinguishes raw GPU forward-pass latency from end-to-end feasible-solution time and states that operational use requires independent verification and feasibility restoration.

Thank you for your consideration.

Sincerely,

Roshan Kumar Sharma
Ujjwal Prasad Dahal
Department of Electrical Engineering
Purwanchal Campus, Institute of Engineering
Tribhuvan University, Dharan, Nepal